L3a: Equity Exchanges, Growth Rates, and Stylized Facts

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we connect equity prices, growth rates, and the patterns a useful market model should reproduce.

Today’s concepts

- Equity securities: explain what stocks and ETFs represent and how they trade.
- Growth rates and log returns: compute both from prices and state their units.
- Stylized facts: identify heavy tails, weak growth-rate autocorrelation, and volatility clustering.

Lecture: Equity Exchanges, Growth Rates, and Stylized Facts

Lectures and Examples

Lecture: Equity Exchanges, Growth Rates, and Stylized Facts

Example: SVD of S&P 500 Growth Rates
Find common patterns across firms and reconstruct the growth-rate matrix using its leading modes.

Example: Stylized Facts in Daily Growth Rates
Compare distributions, estimate tail indices, and compare the ACF of growth rates with the ACF of their magnitudes.

Exchanges, Stocks, and ETFs

Before measuring an equity return, separate the marketplace from the securities traded there.

- An exchange is an organized venue that matches orders to buy and sell securities.
- An order is an investor's instruction to buy or sell a security. Order types differ in how they set the price.
- A stock is a residual ownership claim on a corporation.
- An exchange-traded fund, or ETF, is a security whose shares represent an interest in a portfolio and trade during the day.

Today: a market price is the price available at a particular time and trading venue.

Exchanges Connect Buyers and Sellers

Exchanges match buy and sell orders. Trading helps establish prices (price discovery) and lets investors trade with limited price impact (liquidity).

- Exchanges set listing standards and trading rules.
- NYSE, Nasdaq, and Cboe each operate multiple venues. Cboe BZX is one venue within the Cboe group.
- Orders can be routed to different venues based on prices, available quantities, and order instructions.

Source: SEC registered exchanges. Clearing and settlement transfer the securities and funds after a trade.

Orders Are Instructions to Trade

An order states what to trade, whether to buy or sell, how much, and any price limits.

- **Market order:** seeks an immediate trade at available prices. A large order may trade at several prices.
- **Limit order:** buys at the limit price or lower, or sells at the limit price or higher. It may not execute.
- **Stop and stop-limit orders:** become market or limit orders when the price reaches a trigger.

Source: Investor.gov, Types of Orders

Order Books and the NBBO

An order book holds outstanding buy and sell limit orders. Each venue keeps its own book.

- Bid: the highest displayed buy price. Ask: the lowest displayed sell price. The spread is ask minus bid.
- A market buy begins at the best ask; a market sell begins at the best bid. Larger orders may use deeper price levels.
- The National Best Bid and Offer (NBBO) gives the highest displayed bid and lowest displayed ask across the national market system.

Order protection and best execution: Regulation NMS and FINRA Rule 5310. The NBBO is a quote benchmark.

Order Books and the NBBO for Stock XYZ

Order books and the NBBO

National Best Bid and Offer

NASDAQ • XYZ

SELL / ASK	SIZE	PRICE
	200	50.07
	400	50.05
BEST	100	**50.04**

spread = $0.03

BUY / BID		
BEST	300	**50.01**
	500	50.00
	200	49.99

TOP OF BOOK • XYZ

VENUE	BID	ASK	SPREAD
NYSE	50.00	50.05	0.05
Nasdaq	50.01	50.04	0.03
Cboe BZX	50.02	50.06	0.04
IEX	50.01	50.06	0.05

best bid and best ask sit
on different venues

NBBO • XYZ

BEST BID	BEST ASK
$50.02	**$50.04**
Cboe BZX	Nasdaq

spread $0.02

tighter than any single
venue's own spread

Stocks Are Claims on Firms

A stock, also called a share or equity security, represents fractional ownership in a corporation. It is a residual claim, not a promise of fixed repayment.

What the shareholder may receive

- Voting rights and cash distributions such as dividends.
- A claim on remaining assets only after contractual creditors are paid.
- A capital gain if another investor later pays a higher share price.

Limited liability ordinarily caps the shareholder's loss at the invested amount. It does not protect the share price from falling to zero.

Source: U.S. Securities and Exchange Commission, Stocks

ETFs Are Claims on Portfolios

An ETF holds a portfolio and issues shares that trade throughout the day.

- Exposure: broad-market funds such as SPY hold hundreds of stocks. Sector and single-stock ETFs are more concentrated.
- Leveraged and inverse funds: use derivatives to target a multiple or the negative of a stock's or index's daily move.
- Net asset value (NAV): assets minus liabilities, divided by shares outstanding. The market price can be above or below NAV.

An ETF's risk depends on what it holds and how it invests.

Reward

We now turn from how shares trade to how their prices change. We measure that change with the continuously compounded growth rate, g. Let $S_{j-1}^{(i)} > 0$ and $S_{j}^{(i)} > 0$ be consecutive prices for firm i, $\Delta t_j > 0$ years apart. The next price is given by:

$$S_{j}^{(i)} = S_{j-1}^{(i)} \exp\left(g_{j,j-1}^{(i)} \Delta t_j\right).$$

Solving for the growth rate gives the expression:

$$g_{j,j-1}^{(i)} = \underbrace{\frac{1}{\Delta t_j} \log\left(\frac{S_{j}^{(i)}}{S_{j-1}^{(i)}}\right)}_{\text{growth rate}}.$$

The growth rate has units of inverse years. Its sign records whether the price increased, decreased, or remained unchanged. The notebook computes g from closing-price series, so dividends and other cash distributions are not included.

Growth Rates and Log Returns

Let g_t be the continuously compounded growth rate over an interval of length $\Delta t_t > 0$ years. The corresponding log return r_t is dimensionless and is given by:

$$\underbrace{r_t = (\Delta t_t) g_t}_{\text{log return}} = \log\left(\frac{S_t}{S_{t-1}}\right).$$

Log returns add across adjacent intervals. For positive prices $S_1, \ldots, S_T$, the cumulative log return is given by:

$$\sum_{t=2}^{T} r_t = \underbrace{\sum_{t=2}^{T} (\Delta t_t) g_t}_{\text{time-weighted growth}} = \log\left(\frac{S_T}{S_1}\right).$$

Use g to compare annualized rates. Convert to r before aggregating across time.

Excess Growth Rate

To compare performance with a benchmark, subtract its growth rate.
Let g_t be the equity growth rate and g_y the benchmark growth rate, both in inverse years. Over Δt years, the excess growth rate and excess log return are given by:

$$\underbrace{\bar{g}_t = g_t - g_y}_{\text{excess growth rate}} \quad , \qquad \underbrace{\bar{r}_t = (\Delta t)\bar{g}_t}_{\text{excess log return}} \quad .$$

- $\bar{g}_t > 0$: the equity grew faster than the benchmark.
- $\bar{g}_t < 0$: the equity grew more slowly.
- $\bar{g}_t = 0$: their growth rates were equal.

Comparison: use the same currency, observation interval, and time units.

Algorithm

Given prices $\mathcal{D}_i = \{S_1^{(i)}, \dots, S_T^{(i)}\}$ for firm i, an interval $\Delta t > 0$ years, and benchmark rate g_y in inverse years:

1. For each $t = 2, \dots, T$, read consecutive prices $S_{t-1}^{(i)}$ and $S_t^{(i)}$.
2. Compute the excess growth rate using:

$$\bar{g}_t^{(i)} \leftarrow \underbrace{\frac{1}{\Delta t} \ln\left(\frac{S_t^{(i)}}{S_{t-1}^{(i)}}\right)}_{g_t^{(i)}} - g_y.$$

3. Store the result. The T prices give $T-1$ growth rates.

Set $g_y = 0$ to compute ordinary growth rates.

Common Patterns Across Many Assets

Computing growth rates across firms lets us look for shared patterns. The matrix $\mathbf{G} \in \mathbb{R}^{m \times n}$ holds the observed growth rates for m trading-day intervals and n firms. In the singular value decomposition (SVD), columns $\mathbf{u}_j$ of $\mathbf{U}$ describe time patterns. Columns $\mathbf{v}_j$ of $\mathbf{V}$ give firm weights. The diagonal entries σ_j of Σ give the size of each pattern. Together, these give the decomposition:

$$\mathbf{G} = \mathbf{U}\Sigma\mathbf{V}^\top = \sum_{j=1}^{\mathrm{rank}(\mathbf{G})} \underbrace{\sigma_j \mathbf{u}_j \mathbf{v}_j^\top}_{\text{one mode}}, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$$

Keeping the leading modes gives a simpler approximation of the matrix.

Example: SVD of S&P 500 Growth Rates. Shared patterns do not by themselves explain economic causes.

Risk

Shared patterns describe how firms move together. For each firm, volatility measures how widely growth rates vary around their mean.

Let $g_2, \ldots, g_T$ be the $T-1$ growth rates from T equally spaced prices. Their mean g' and sample volatility σ_g are given by:

$$g' = \frac{1}{T-1}\sum_{t=2}^{T} g_t, \qquad \sigma_g = \underbrace{\sqrt{\frac{1}{T-2}\sum_{t=2}^{T}(g_t - g')^2}}_{\text{growth-rate volatility}}.$$

Both g_t and σ_g have units of inverse years. Volatility describes spread; it does not describe every kind of risk.

Log-Return and Annualized Volatility

For a fixed interval Δt years, the log return is $r_t = (\Delta t) g_t$. Its standard deviation $\sigma_r(\Delta t)$ is given by:

$$\underbrace{\sigma_r(\Delta t) = (\Delta t)\sigma_g}_{\text{interval log-return volatility}} \quad .$$

This is an exact change of units. The result is dimensionless.
The conventional annualized value σ_{ann} is given by:

$$\underbrace{\sigma_{\mathrm{ann}} = \frac{\sigma_r(\Delta t)}{\sqrt{\Delta t}} = \sqrt{\Delta t}\,\sigma_g}_{\text{annualized log-return volatility}} \qquad [\mathrm{yr}^{-1/2}].$$

Assumption: interpreting this annualized value across horizons assumes uncorrelated returns with stable variance. It is not an exact one-year forecast.

Other Risk Measures

Choose a measure that fits the loss you care about.

- Large losses: Value at Risk sets a loss threshold for a stated probability and horizon. Expected Shortfall averages losses beyond that threshold.
- Shortfalls and drawdowns: downside deviation measures shortfalls below a target. Maximum drawdown measures the largest fall from a past peak.
- Benchmark risk: beta measures sensitivity to market returns. Tracking error is the standard deviation of returns minus benchmark returns.

Stylized Facts of Equity Growth Rates

Volatility summarizes spread, but not the full shape or timing of price movements. A stylized fact is a recurring pattern across assets and time periods. We focus on three:

- Heavy tails: large growth rates occur more often than a fitted Normal model predicts.
- Weak linear autocorrelation: growth rates on different days usually have little linear relationship beyond short lags.
- Volatility clustering: large movements tend to occur together, as do small movements.

Sources: Cont (2001) and Mandelbrot (1963)

Heavy-Tailed Growth-Rate Distributions

Large positive and negative growth rates occur more often than a Normal model predicts. Three distributions give useful comparisons:

- Normal: thin tails; our benchmark.
- Laplace: a sharper center and heavier tails than Normal, but the tails still decrease exponentially.
- Student-t: power-law tails. Smaller degrees of freedom ν give heavier tails. The true tail index is $\alpha = \nu$.

Laplace can fit the center well. Check the far tails before choosing a model for extreme events.

Power-Law Tails

Let $X = |g|$ be the growth-rate magnitude and $\alpha > 0$ the tail index. For a regularly varying tail, the tail probability follows the asymptotic form:

$$\mathbb{P}(X > x) \sim \underbrace{L(x)x^{-\alpha}}_{\text{tail probability}} \qquad \text{as } x \to \infty.$$

The symbol $\sim$ means the ratio of the two sides approaches one. The factor $L(x)$ is slowly varying, meaning that for every fixed $c > 0$, the following limit holds:

$$\lim_{x\to\infty} \frac{L(cx)}{L(x)} = 1.$$

A smaller α means heavier tails and more extreme events. Student-t tails follow this form; Normal and Laplace tails do not.

Why the Tail Index Matters

Smaller estimated α means heavier tails and more extreme events.

We simulated 5,000 samples per distribution, each with 2,766 observations, and used the 52 largest absolute movements to estimate α.

Distribution	Median estimate	Range containing 90%
Normal	7.8	6.5–9.7
Laplace	4.8	4.0–6.0
Student-t, $\nu = 4$	3.5	2.9–4.4
Student-t, $\nu = 3$	2.8	2.3–3.6

Simulation: independent observations, mean zero, equal variance. Ranges depend on sample size and cutoff. Overlap means an estimate alone cannot identify the distribution.

True and Estimated Tail Indices

For Student-t, the true tail index is $\alpha = \nu$. The value can be fractional.

- True index: a Student-t model with $\nu = 4$ has $\alpha = 4$.
- Estimated index: our simulation gives a median Hill estimate of $\hat{\alpha} = 3.5$ for that same model.
- Estimates depend on the sample and how many extreme observations we include.

Example: the fitted Student-t parameter uses all observations. Hill uses only the largest magnitudes. Their estimates can differ.

Hill Estimator of the Tail Index

Sort the n positive growth-rate magnitudes from largest to smallest: $X_{(1)} \geq \cdots \geq X_{(n)} > 0$. Choose $k < n$ magnitudes for the tail. The next value, $X_{(k+1)}$, sets the threshold. The Hill estimate is given by:

$$\hat{\alpha}_k = \underbrace{\left[\frac{1}{k}\sum_{i=1}^{k}\ln\left(\frac{X_{(i)}}{X_{(k+1)}}\right)\right]^{-1}}_{\text{estimated tail index}}.$$

Too few observations give a noisy estimate. Too many include values outside the far tail.

Check: plot $\widehat{\alpha}_k$ across several k values and look for a stable region. Hill estimation assumes a regularly varying tail.

Autocorrelation

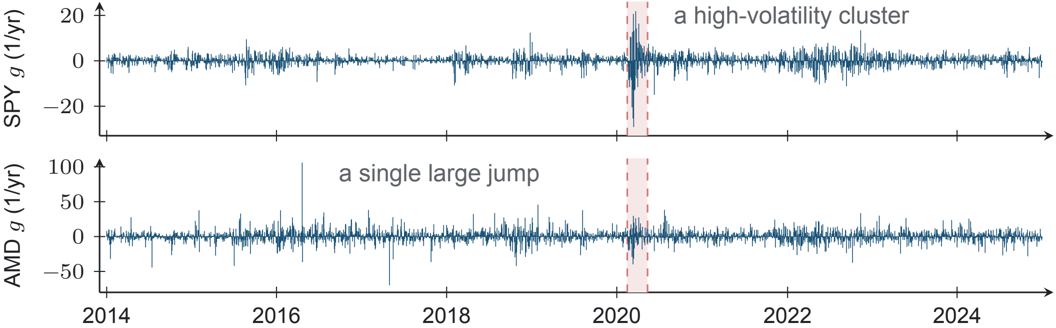
We have measured movement sizes. Next, we ask how they relate across time. For growth rates $g_2, \dots, g_T$ with sample mean g', let k be the lag in time steps. The empirical autocovariance and autocorrelation function (ACF) are given by:

$$\widehat{R}(k) = \frac{1}{T-1} \sum_{t=2}^{T-k} (g_t - g')(g_{t+k} - g'), \qquad \underbrace{\rho(k) = \frac{\widehat{R}(k)}{\widehat{R}(0)}}_{\text{autocorrelation}}.$$

- $\rho(0) = 1$ and $|\rho(k)| \leq 1$.
- $\rho(k)$ near zero means little linear relationship at that lag.
- Weak growth-rate autocorrelation does not establish independence. Movement sizes may still be related.

Random-walk benchmark: expect growth-rate autocorrelations near zero for $k > 0$.

Volatility Clustering



Volatility clustering: large-magnitude days arrive in bursts of both signs, and the marked weeks are turbulent in both series. One large jump is not a cluster.

Data: daily SPY and AMD growth rates, 2014–2024, from the stylized-facts example dataset.

Diagnosing Volatility Clustering

Repeat the ACF calculation using magnitudes. For firm i, let $\overline{|g|}_i$ be the mean of $|g_2^{(i)}|, \dots, |g_T^{(i)}|$. The magnitude ACF at lag k is given by:

$$\hat{\rho}_i(k) = \frac{\displaystyle\sum_{t=2}^{T-k} \left[|g_t^{(i)}| - \overline{|g|}_i\right] \left[|g_{t+k}^{(i)}| - \overline{|g|}_i\right]}{\displaystyle\sum_{t=2}^{T} \left[|g_t^{(i)}| - \overline{|g|}_i\right]^2}.$$

Positive magnitude autocorrelations across many lags are evidence of volatility clustering.

Main comparison: growth-rate autocorrelations can be small while magnitude autocorrelations remain positive.

Example: Stylized Facts

We can now check all three patterns in daily growth-rate data.

- Compare observed growth rates with Normal, Laplace, and Student-t distributions.
- Estimate the tail index across several cutoffs.
- Compare the ACF of growth rates with the ACF of their magnitudes.

Example: Stylized Facts in Daily Growth Rates

Optional Advanced Material

These examples extend the market-structure discussion.

- Limit-Order Book Mechanics
  Match orders, process cancellations, and calculate the prices paid by a market order.

- Market Impact Models
  Compare the cost of using the order book with models of how trades affect prices.

Suggested order: complete the order-book example first. Neither is required for L3b.

Summary

This lecture connected equity trading, price growth, and patterns in market data.

Key takeaways

- **Trading determines the price.** Order instructions and available quotes affect where and how a trade fills.
- **State the units and horizon.** Growth rates, log returns, and their volatilities describe the same price changes on different scales.
- **Check all three patterns.** Growth rates have heavy tails and weak linear autocorrelation; their magnitudes remain correlated over time.

Next, we test a binomial lattice against these patterns.

That’s a wrap. What are some of the interesting things we discussed today?